

Name: _____ Date: _____

Period: _____

Problems 1–6 — Key

1. Unit circle evaluations.

(a) $\tan(0) = 0$

(b) $\tan\left(\frac{\pi}{4}\right) = 1$

(c) $\tan\left(\frac{\pi}{2}\right) = \text{undefined}$

(d) $\tan(\pi) = 0$

(e) $\tan\left(\frac{3\pi}{4}\right) = -1$

(f) $\tan\left(-\frac{\pi}{4}\right) = -1$

2. Periods and asymptotes.

(a) $g(\theta) = \tan\left(\frac{\theta}{3}\right)$

Period = 3π

Asymptotes: $\theta = \frac{3\pi}{2} + 3k\pi$ for integer k

Teacher Note

$b = 1/3$; period = $\pi/(1/3) = 3\pi$. Asymptotes: $\theta/3 = \pi/2 + k\pi \Rightarrow \theta = 3\pi/2 + 3k\pi$.

(b) $h(\theta) = \tan(4\theta)$

Period = $\pi/4$

Asymptotes: $\theta = \frac{\pi}{8} + \frac{k\pi}{4}$ for integer k

(c) $p(\theta) = -3\tan(\theta) + 2$

Period = π

Asymptotes: $\theta = \frac{\pi}{2} + k\pi$ for integer k

Teacher Note

a and d do not affect period or asymptote locations.

3. $f(\theta) = \tan\left(\theta - \frac{\pi}{6}\right)$

(a) Phase shift: $+\pi/6$ (shift right $\pi/6$ units)

(b) Asymptote closest to $\theta = 0$ with $\theta > 0$:

Set $\theta - \pi/6 = \pi/2 \Rightarrow \theta = \pi/2 + \pi/6 = 3\pi/6 + \pi/6 = 4\pi/6 = 2\pi/3$. The asymptote is at $\theta = 2\pi/3$.

4. Description:

The graph of $g(\theta) = 3\tan(2\theta)$ has period $\pi/2$ (half the period of $\tan\theta$) and all output values are tripled (vertical dilation by 3). The asymptotes are twice as close together, and the graph rises and falls three times as steeply near each inflection point.

5. AP-Style MC.

Teacher Note

Set $\frac{\theta}{2} + \frac{\pi}{4} = \frac{\pi}{2} + k\pi$, so $\frac{\theta}{2} = \frac{\pi}{4} + k\pi$, so $\theta = \frac{\pi}{2} + 2k\pi$. For $k = 0$: $\theta = \pi/2$ (B); for $k = 1$: $\theta = 5\pi/2$; etc.

(A) $\theta = \frac{\pi}{4}$

(B) $\theta = \frac{\pi}{2}$ ← correct

(C) $\theta = \pi$

(D) $\theta = \frac{3\pi}{2}$

(E) $\theta = 2\pi$

Answer: **B**

6.

(a) **Yes.** $\tan(-\pi/4) = -1$, $\tan(0) = 0$, $\tan(\pi/4) = 1$, $\tan(\pi/2)$ is undefined, $\tan(3\pi/4) = -1$. All values match the definition $\tan \theta = \sin \theta / \cos \theta$.

(b) **The period suggested by the table is π (the pattern from $-\pi/4$ to $3\pi/4$ repeats every π units).**

Extension optional

Extension (Optional) — Key:

$$\tan(\theta + \pi) = \frac{\sin(\theta + \pi)}{\cos(\theta + \pi)} = \frac{-\sin \theta}{-\cos \theta} = \frac{\sin \theta}{\cos \theta} = \tan \theta.$$

The negatives cancel, confirming $\tan(\theta + \pi) = \tan \theta$. This is why the period is π .

Preview — Lesson 3.9: Inverse Trigonometric Functions

In Lesson 3.9 we will ask: if we know a sine, cosine, or tangent value, can we find the angle? To answer that question we restrict the domain of each trigonometric function to make it invertible, then define arcsin, arccos, and arctan. The asymptote behavior you studied today — $\tan \theta \rightarrow \pm\infty$ as $\theta \rightarrow \pm\pi/2$ — is precisely what tells us the range of arctan.